§1 σ-algebra manipulation

Problem 1.1 (Unions and Intersections) — Let E be a set and let S be a set of σ-algebras on E. Define

$$\mathcal{E}^* = \{A \subseteq E : A \in \mathcal{E} \text{ for all } \mathcal{E} \in S\}.$$

Show that $\mathcal{E}^*$ is a σ-algebra on E. Show, on the other hand, by example, that the union of two σ-algebras on the same set need not be a σ-algebra.

Just check the axioms:

(a) $\emptyset \in \mathcal{E}$ for all $\mathcal{E} \in S$, so $\emptyset \in \mathcal{E}^*$.

(b) If $A \in \mathcal{E}^*$, then $A \in \mathcal{E}$ for all $\mathcal{E} \in S$. But then $A^c \in \mathcal{E}$ for all $\mathcal{E} \in S$, so $A^c \in \mathcal{E}^*$.

(c) If $(A_n : n \in \mathbb{N})$ is a sequence of sets in $\mathcal{E}^*$, then $A_n \in \mathcal{E}$ for all $\mathcal{E} \in S$ and all $n \in \mathbb{N}$. But then $\bigcup_n A_n \in \mathcal{E}$ for all $\mathcal{E} \in S$, so $\bigcup_n A_n \in \mathcal{E}^*$.

The counterexample is really stupid: let $E = \{1, 2, 3, 4\}$, let $\mathcal{E}_1 = \{\emptyset, \{1, 2\}, \{3, 4\}, E\}$, and let $\mathcal{E}_2 = \{\emptyset, \{1, 3\}, \{2, 4\}, E\}$. Then $\mathcal{E}_1 \cup \mathcal{E}_2 = \{\emptyset, \{1, 2\}, \{1, 3\}, \{2, 4\}, \{3, 4\}, E\}$ is not closed under countable unions (where is $\{1, 2, 3\}$?).

§2 Real number computations

Problem 2.1 — Show that the following sets of subsets of $\mathbb{R}$ all generate the same σ-algebra:

(a) $\{(a, b) : a < b\}$,

(b) $\{(a, b] : a < b\}$,

(c) $\{(-\infty, b] : b \in \mathbb{R}\}$.

For notation, let $\mathcal{E}_a, \mathcal{E}_b, \mathcal{E}_c$ denote the σ-algebras generated by $(a), (b), (c)$ respectively.

(Proof $\mathcal{E}_c \subset \mathcal{E}_a$) By complements, $\mathcal{E}_a$ contains all sets of the form

$$(a, b]^c = (-\infty, a] \cup [b, \infty)$$

for all $a < b$. By countable intersections, we have

$$\bigcup_{b \in \mathbb{Z}, b > a} (-\infty, a] \cup [b, \infty) = (-\infty, a]$$

thus $\mathcal{E}_c \subset \mathcal{E}_a$.

(Proof $\mathcal{E}_a \subset \mathcal{E}_b$) By countable unions, $\mathcal{E}_b$ contains all sets of the form

$$\bigcup_{n \in \mathbb{N}} (a, b - 1/n] = (a, b)$$

for all $a < b$. Thus $\mathcal{E}_a \subset \mathcal{E}_b$.

(Proof $\mathcal{E}_b \subset \mathcal{E}_c$) By complements and intersections, $\mathcal{E}_c$ contains all sets of the form

$$(-\infty, b] \setminus (-\infty, a] = (a, b]$$

for all $a < b$. Thus $\mathcal{E}_b \subset \mathcal{E}_c$.

Problem 2.2 — Show that a countably additive set function on a ring $\mathcal{S}$ is additive, increasing and countably subadditive.

Countably additive sets are automatically additive. This is also clearly increasing because for any $A, B \in \mathcal{S}$ such that $A \subset B$, $B \setminus A \in \mathcal{S}$, thus

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

Finally, we show countable subadditivity. For all sequences $A_1, A_2, \cdots \in \mathcal{S}$ such that $\bigcup_n A_n \in \mathcal{S}$, we know that $B_i = A_i \setminus \bigcup_{n<i} A_n \in \mathcal{S}$ for all $i \geq 1$, and of course

$$\bigcup_n A_n = \bigcup_n B_n.$$

On the other hand, the B_n are disjoint by construction, so

$$\mu\left(\bigcup_n A_n\right) = \mu\left(\bigcup_n B_n\right) = \sum_n \mu(B_n) \leq \sum_n \mu(A_n).$$

where the last statement is because $B_n \subset A_n$ and we know μ is increasing.

§3 Dynkin systems

Problem 3.1 — Show that a π-system which is also a d-system is a σ-algebra.

First off, all σ-algebras are clearly π systems. They are d-systems, because:

(a) Ω is in the σ-algebra.

(b) If $A \subset B$ are in the σ-algebra, then $B \setminus A = B \cap A^c$ is also in the σ-algebra.

(c) If $A_1, A_2, \cdots$ are in the σ-algebra, then $\bigcup_n A_n$ is also in the σ-algebra (This is stronger than we need; a d-system needs only be closed under countable increasing union).

If you are a d system, then you have Ω. Thus by the existence of proper differences you have complements. If you also a π system, then finite intersections and complements give you finite unions via De Morgan's law.

Finally, given any collection $A_1, A_2, \cdots \in \mathcal{L}$, their prefix unions $A_1 \cup \cdots A_n$ are all in $\mathcal{L}$, and thus by the countable increasing union condition of d-systems, their union is also in $\mathcal{L}$.

Problem 3.2 (Countable Additivity Condition) — Let μ be a finite-valued additive set function on a ring $\mathcal{A}$. Show that μ is countably additive if and only if the following condition holds: for any decreasing sequence $(A_n : n \in \mathbb{N})$ of sets in $\mathcal{A}$, with $\bigcap_n A_n = \emptyset$, we have $\mu(A_n) \to 0$ as $n \to \infty$.

(Proof $\Rightarrow$) Suppose μ is countably additive. In particular we're additive, so we can let $B_i = A_i \setminus A_{i+1} \in \mathcal{A}$ for all $i \geq 1$; these are all disjoint by construction. Then by countable additivity, we know that

$$\infty > \mu(A_n) = \mu\left(\bigcup_{i \geq n} B_i\right) = \sum_{i \geq n} \mu(B_i)$$

for all $n \geq 1$. By definition of infinite sums, for all $\varepsilon > 0$, there exists an M such that

$$\sum_{n \leq i \leq n+M} \mu(B_i) > \sum_{i \geq n} \mu(B_i) - \varepsilon \implies \sum_{n+M \leq i} \mu(B_i) < \varepsilon.$$

Thus $\mu(A_n) \to 0$ as $n \to \infty$.

(Proof $\Leftarrow$) Suppose we know the second condition. Then, for any collection of disjoint sets $A_1, A_2, \cdots \in \mathcal{A}$ such that $A = \bigcup_n A_n \in \mathcal{A}$, we can define, we can define

$$B_n = \bigcup_{i \geq n} A_i = A \setminus \left(\bigcup_{i < n} A_i\right) \in \mathcal{A}$$

Clearly, B_n form a decreasing sequence such that $\bigcap_n B_n = \emptyset$, thus $\mu(B_n) \to 0$ as $n \to \infty$.

But this is what we want. First off, for any ε, there exists an N such that $\mu(B_N) < \varepsilon$, thus

$$\varepsilon + \sum_{i \in \mathbb{N}} \mu(A_n) \geq \mu(B_N) + \sum_{i < N} \mu(A_n) = \mu(A)$$

thus $\sum_{i \in \mathbb{N}} \mu(A_n) \geq \mu(A)$. On the other hand, for all N,

$$\sum_{i \leq N} \mu(A_i) = \mu\left(\bigcup_{i \leq N} A_i\right) \leq \mu(A)$$

(the inequality is because additive set functions on rings are automatically increasing), thus we can pass to the limit.

4 Interesting things

Problem 4.1 (Baby Fatou-Lebesgue Theorem) — Let $(E, \mathcal{E}, \mu)$ be a finite measure

space. Show that, for any sequence of sets $(A_n : n \in \mathbb{N})$ in $\mathcal{E}$,

$$\mu\left(\liminf_n A_n\right) \leq \liminf_n \mu(A_n) \leq \limsup_n \mu(A_n) \leq \mu\left(\limsup_n A_n\right).$$

Show that the first inequality remains true without the assumption that $\mu(E) < \infty$, but that the last inequality may then be false.

The middle inequality is true because for any set S, $\inf S \leq \sup S$.

Claim (Continuity from Above and Below) — We write $A_i \uparrow A$ if $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_i A_i$. Then,

$$\mu(A) = \lim_i \mu(A_i).$$

Similarly, we write $A_i \downarrow A$ if $A_1 \supseteq A_2 \supseteq \dots$ and $A = \bigcap_i A_i$. If $\mu(A_1) < \infty$, then

$$\mu(A) = \lim_i \mu(A_i).$$

Proof. Let $A = \bigcup_i (A_i \setminus A_{i-1})$, each of which is disjoint. Then

$$\mu(A) = \sum_i \mu(A_i \setminus A_{i-1}) = \lim_i \sum_{j \leq i} \mu(A_j \setminus A_{j-1}) = \lim_i \mu(A_i).$$

For the $\downarrow$ direction, apply the previous result to $A_1 \setminus A_i$ to yield

$$\mu(A_1 \setminus A) = \lim_i \mu(A_1 \setminus A_i) = \lim_i \mu(A_1) - \mu(A_i) = \mu(A_1) - \lim_i \mu(A_i)$$

We need $\infty > \mu(A_1) > \mu(A_i)$ for all the subtractions to avoid $\infty - \infty$. □

(First inequality) We do not assume $\mu(E) < \infty$. First we construct

$$B_n = \bigcap_{k \geq n} A_k \in \mathcal{E}$$

(note that these are increasing; this is the usual $\sup\inf = \liminf$ identity). Now the claim applies, so we must show

$$\lim_{n\to\infty} \mu(B_n) \leq \lim_{n\to\infty} \inf_{k \geq n} \mu(A_k)$$

Actually, this is just true termwise:

$$\mu\left(\bigcap_{k \geq n} A_k\right) \leq \mu(A_i)$$

for all $i \geq n$, thus

$$\mu\left(\bigcap_{k \geq n} A_k\right) \leq \inf_{k \geq n} \mu(A_k)$$

(Third inequality) Do the same thing.

$$B_n = \bigcup_{k \geq n} A_k \in \mathcal{E}$$

These are decreasing and $\mu(B_1) < \mu(E) < \infty$, so we apply the claim: we want

$$\lim_{n\to\infty} \mu(B_n) \geq \lim_{n\to\infty} \sup_{k\geq n} \mu(A_k)$$

This is again true termwise:

$$\mu\left(\bigcup_{k\geq n} A_k\right) \geq \mu(A_i)$$

for all $i \geq n$, thus

$$\mu\left(\bigcup_{k\geq n} A_k\right) \geq \sup_{k\geq n} \mu(A_k)$$

Problem 4.2 — Let $(A_n : n \in \mathbb{N})$ be a sequence of events in a probability space Ω. Show that the events A_n are independent if and only if the σ-algebras $\sigma(A_n) = \{\emptyset, A_n, A_n^c, \Omega\}$ are independent.

If the σ-algebras are independent, then by definition for any $J \subset \mathbb{N}$,

$$\mu\left(\bigcap_{i\in J} A_i\right) = \prod_{i\in J} \mu(A_i)$$

thus the events are independent.

We now want a stupid claim: if $(A_n : n \leq N)$ are a finite sequence of independent events, then $\{A_1^c, A_2, A_3, \cdots A_n\}$ (where we have A_1^c instead of A_1) are also independent. This is clear; if $J \subset \{2, \ldots, N\}$, then nothing has changed; if $J = \{1\} \cup K$ with $K \subset \{2, \ldots, N\}$, then

$$\begin{aligned}
\mu\left(A_1^c \cap \bigcap_{i\in K} A_i\right) &= \mu\left(\bigcap_{i\in K} A_i\right) - \mu\left(A_1 \cap \bigcap_{i\in K} A_i\right) \\
&= \prod_{i\in K} \mu(A_i) - \mu(A_1) \cdot \prod_{i\in K} \mu(A_i) \\
&= \mu(A_1^c) \cdot \prod_{i\in K} \mu(A_i)
\end{aligned}$$

as desired. By repeated applying this a finite number of times, we find that any finite subcollection of the A_n can be replaced by its complements without changing the independence.

Given the events are independent, lets consider any family of sets $B_n \in \sigma(A_n)$ for $n \in \mathbb{N}$. We want to show

$$\mu\left(\bigcap_{n\in\mathbb{N}} B_n\right) = \prod_{n\in\mathbb{N}} \mu(B_n)$$

By continuity from above and definition of infinite product, it suffices to show

$$\mu\left(\bigcap_{n\leq N} B_n\right) = \prod_{n\leq N} \mu(B_n)$$

for all N. We induct; the base case $n = 0$ is clear.

Then, assume we know the $n-1$ case. If any B_i is $\emptyset$, both sides are 0. If any B_i is Ω, we can remove it from the expression without changing either side, and reduce to the $n-1$ case. Thus we can assume all B_i are nontrivial. By the claim, we are now done.

Problem 4.3 — Let B be a Borel subset of the interval $[0,1]$. Show that for every $\epsilon > 0$, there exists a finite union of disjoint intervals $A = (a_1, b_1] \cup \cdots \cup (a_n, b_n]$ such that the Lebesgue measure of $A \triangle B = (A^c \cap B) \cup (A \cap B^c)$ is less than ϵ. Show further that this remains true for every Borel set in $\mathbb{R}$ of finite Lebesgue measure.

Let $\mathcal{P} \subset \mathcal{B}(\mathbb{R})$ contain all Borel subsets B satisfying the claim. We claim that $\mathcal{P}$ is a π-system. Indeed, for any $C, D \in \mathcal{P}$, for every $\varepsilon > 0$, there exists A and B which are finite unions of disjoint intervals such that

$$\mu\left(A \triangle C\right), \mu\left(B \triangle D\right) < 0.0001\varepsilon$$

Then, $A \cap B$ is a finite union of disjoint intervals.[1] We now bash some inequalities:

$$\begin{aligned}
\mu\left\{(A\cap B)\triangle(C\cap D)\right\} &= \mu\left\{((A\cap B)\cap(C^c\cup D^c))\cup((C\cap D)\cap(A^c\cup B^c))\right\} \\
&= \mu\left\{((A\cap B)\cap C^c)\cup((A\cap B)\cap D^c)\right. \\
&\left.\cup((C\cap D)\cap A^c)\cup((C\cap D)\cap B^c)\right\} \\
&\leq \mu\left((A\cap B)\cap C^c\right) + \mu\left((A\cap B)\cap D^c\right) \\
&+ \mu\left((C\cap D)\cap A^c\right) + \mu\left((C\cap D)\cap B^c\right) \\
&\leq \mu\left(A\cap C^c\right) + \mu\left(B\cap D^d\right) + \mu\left(C\cap A^c\right) + \mu\left(D\cap B^c\right) \\
&< 0.0002\varepsilon
\end{aligned}$$

5 Caratheodory yields complete measures

Problem 5.1 — Let $(E, \mathcal{E}, \mu)$ be a measure space. Call a subset $N \subseteq E$ null if $N \subseteq B$ for some $B \in \mathcal{E}$ with $\mu(B) = 0$. Write $\mathcal{N}$ for the set of null sets. Prove that the set of subsets

$$\mathcal{E}_\mu = \{A \cup N : A \in \mathcal{E}, N \in \mathcal{N}\}$$

is a σ-algebra and show that μ has a well-defined and countably additive extension to $\mathcal{E}_\mu$ given by $\mu(A \cup N) = \mu(A)$. We call $\mathcal{E}_\mu$ the completion of $\mathcal{E}$ with respect to μ.

Suppose now that $\mathcal{E}$ is σ-finite and write μ^* for the outer measure associated to μ, as in the proof of Caratheodory's extension theorem. Show that $\mathcal{E}_\mu$ is exactly the set of μ^*-measurable sets.

[1] Okay we can verify this; if $A = E_1 \cup \cdots \cup E_n$ and $B = F_1 \cup \cdots \cup F_m$ are finite unions of disjoint intervals, then $A \cap B = E_1 \cap F_1 \cup \cdots \cup E_n \cap F_m$ is a finite union of disjoint intervals, some of which might be empty.

(Proof $\mathcal{E}_\mu$ is a σ-algebra) We check the axioms:

(a) $\emptyset \in \mathcal{E}$ and $\emptyset \in \mathcal{N}$, so $\emptyset \in \mathcal{E}_\mu$.

(b) In the above definition where $N \subset B \in \mathcal{E}$ with $\mu(B) = 0$, note that $B \setminus N \subset B$ too. This lets us write

$$(A \cup N)^c = A^c \setminus N = \underbrace{A^c \setminus B}_{\in \mathcal{E}} \cap \underbrace{B \setminus N}_{\in \mathcal{N}}$$

thus $(A \cup N)^c \in \mathcal{E}_\mu$.

(c) If $(A_n \cup N_n : n \in \mathbb{N})$ is a sequence of sets in $\mathcal{E}_\mu$, with each $N_n \subset B_n$ with $\mu(B_n) = 0$, then

$$\bigcup_n (A_n \cup N_n) = \underbrace{\bigcup_n A_n}_{\in \mathcal{E}} \cup \underbrace{\bigcup_n N_n}_{\subset \bigcup_n B_n}$$

Note that $\mu\left(\bigcup_n B_n\right) \leq \sum_n \mu(B_n) = 0$, so $\bigcup_n (A_n \cup N_n) \in \mathcal{E}_\mu$.

(Extension) To show this is well-defined, we need to show that the relation $A \sim C$ iff there exists $N, M \in \mathcal{N}$ such that $A \cup N = C \cup M$ is an equivalence relation. Reflexivity and symmetry are clear. For transitivity, if $A \cup N = C \cup M$ and $C \cup P = D \cup Q$, then

$$A \cup (N \cup P) = C \cup M \cup P = D \cup (Q \cup M)$$

where $N \cup P$ and $Q \cup M$ are both null, so $A \sim D$.

(Countable Additivity) Let $(A_n \cup N_n : n \in \mathbb{N})$ be a sequence of disjoint sets in $\mathcal{E}_\mu$. Then

$$\begin{aligned}
\mu\left(\bigcup_n (A_n \cup N_n)\right) &= \mu\left(\bigcup_n A_n \cup \bigcup_n N_n\right) \\
&= \mu\left(\bigcup_n A_n\right) \\
&= \sum_{n \in \mathbb{N}} \mu(A_n) \\
&= \sum_{n \in \mathbb{N}} \mu(A_n \cup N_n)
\end{aligned}$$

Finish this!

§6 Function manipulation

Problem 6.1 (Preliminaries) — Let $(f_n : n \in \mathbb{N})$ be a sequence of measurable functions on a measurable space $(E, \mathcal{E})$. Show that the following functions are also measurable:

(a) $f_1 + f_2$,

(b) $f_1 f_2$,

(c) $\inf_n f_n$,

(d) $\sup_n f_n$,

(e) $\liminf_n f_n$,

(f) $\limsup_n f_n$.

Show also that $\{x \in E : f_n(x) \text{ converges as } n \to \infty\} \in \mathcal{E}$.

Problem 6.2 (Pushforward measure) — Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measurable spaces, let μ be a measure on E, and let $f : E \to G$ be a measurable function. Show that we can define a measure ν on G by setting $\nu(A) = \mu(f^{-1}(A))$ for each $A \in \mathcal{G}$.

7 Independence

Problem 7.1 — Show that the following condition implies that the random variables X and Y are independent: $\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x)\mathbb{P}(Y \leq y)$ for all $x, y \in \mathbb{R}$.